# Fine-Tuning BERT on a Kaggle Dataset

### ASSIGNMENT NLP – 4

## Objective

The objective of this assignment is to build a text classification model by fine-tuning a pre-trained BERT model on a Kaggle dataset. The process includes preprocessing text data, tokenization, training the model, evaluating performance, and comparing results from different experiments.

# IMPORT LIBRARIES

```
import pandas as pd
import numpy as np
import re
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, precision_recall_fscore_support, cc

import torch
from torch.utils.data import Dataset

from transformers import AutoTokenizer, AutoModelForSequenceClassification, Tra
```

# Load Dataset

In this assignment, IMDB Movie Reviews dataset is used for sentiment classification. It contains text reviews labeled as positive or negative.

```
df = pd.read_csv("IMDB Dataset.csv")
df = df.sample(10000, random_state=42)
df.head()
```

| | review | sentiment |
|---|---|---|
| **33553** | I really liked this Summerslam due to the look... | positive |
| **9427** | Not many television shows appeal to quite as m... | positive |
| **199** | The film quickly gets to a major chase scene w... | negative |
| **12447** | Jane Austen would definitely approve of this o... | positive |
| **39489** | Expectations were somewhat high for me when I ... | negative |

Next steps: Generate code with df New interactive sheet

```
df.info()
df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 10000 entries, 33553 to 5857
Data columns (total 2 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   review     10000 non-null  object
 1   sentiment  10000 non-null  object
dtypes: object(2)
memory usage: 234.4+ KB
```

| | 0 |
|---|---|
| **review** | 0 |
| **sentiment** | 0 |

**dtype:** int64

## Data Preprocessing

Text is cleaned by converting to lowercase, removing special characters, and eliminating extra spaces.

```
df = df.dropna()

def clean_text(text):
    text = text.lower()
    text = re.sub(r"http\S+", "", text)
    text = re.sub(r"[^a-zA-Z\s]", "", text)
    text = re.sub(r"\s+", " ", text).strip()
    return text
```

```
df["clean_text"] = df["review"].apply(clean_text)
df.head()
```

| | review | sentiment | clean_text |
|---|---|---|---|
| **33553** | I really liked this Summerslam due to the look... | positive | i really liked this summerslam due to the look... |
| **9427** | Not many television shows appeal to quite as m... | positive | not many television shows appeal to quite as m... |
| **199** | The film quickly gets to a major chase scene w... | negative | the film quickly gets to a major chase scene w... |
| **12447** | Jane Austen would definitely approve of this o... | positive | jane austen would definitely approve of this o... |
| **39489** | Expectations were somewhat high for me when I ... | negative | expectations were somewhat high for me when i ... |

Next steps: Generate code with df New interactive sheet

## LABEL ENCODING

```
label_map = {"positive": 1, "negative": 0}
df["label"] = df["sentiment"].map(label_map)

df[["clean_text", "label"]].head()
```

| | clean_text | label |
|---|---|---|
| **33553** | i really liked this summerslam due to the look... | 1 |
| **9427** | not many television shows appeal to quite as m... | 1 |
| **199** | the film quickly gets to a major chase scene w... | 0 |
| **12447** | jane austen would definitely approve of this o... | 1 |
| **39489** | expectations were somewhat high for me when i ... | 0 |

## Data Splitting

Dataset is split into training, validation, and testing sets.

```
train_texts, temp_texts, train_labels, temp_labels = train_test_split(
    df["clean_text"].tolist(),
    df["label"].tolist(),
    test_size=0.3,
    random_state=42,
    stratify=df["label"]
)

val_texts, test_texts, val_labels, test_labels = train_test_split(
    temp_texts,
    temp_labels,
    test_size=0.5,
    random_state=42,
    stratify=temp_labels
)

print("Train:", len(train_texts))
print("Validation:", len(val_texts))
print("Test:", len(test_texts))
```

```
Train: 7000
Validation: 1500
Test: 1500
```

## Tokenization

Using bert-base-uncased tokenizer to convert text into tokens.

```
tokenizer = AutoTokenizer.from_pretrained("bert-base-uncased")

train_encodings = tokenizer(train_texts, truncation=True, padding=True, max_ler
val_encodings = tokenizer(val_texts, truncation=True, padding=True, max_length=
test_encodings = tokenizer(test_texts, truncation=True, padding=True, max_lengt
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: Userw
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access publ
  warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated req
```

```
config.json: 100%                                    570/570 [00:00<00:00, 60.5kB/s]

tokenizer_config.json: 100%                          48.0/48.0 [00:00<00:00, 5.11kB/s]

vocab.txt:        232k/? [00:00<00:00, 5.64MB/s]

tokenizer.json:        466k/? [00:00<00:00, 7.02MB/s]
```

# DATASET CLASS

```
class TextDataset(Dataset):
    def __init__(self, encodings, labels):
        self.encodings = encodings
        self.labels = labels

    def __getitem__(self, idx):
        item = {key: torch.tensor(val[idx]) for key, val in self.encodings.item
        item["labels"] = torch.tensor(self.labels[idx])
        return item

    def __len__(self):
        return len(self.labels)

train_dataset = TextDataset(train_encodings, train_labels)
val_dataset = TextDataset(val_encodings, val_labels)
test_dataset = TextDataset(test_encodings, test_labels)
```

# METRICS FUNCTION

```
def compute_metrics(eval_pred):
    logits, labels = eval_pred
    preds = np.argmax(logits, axis=1)

    precision, recall, f1, _ = precision_recall_fscore_support(labels, preds, a
    acc = accuracy_score(labels, preds)

    return {
        "accuracy": acc,
```

```
            "precision": precision,
            "recall": recall,
            "f1": f1
        }
```

## Experiment 1: Freeze BERT Layers

In this experiment, all BERT layers are frozen and only the classifier layer is trained. This reduces training time but limits the model's ability to learn task-specific features.

```
model1 = AutoModelForSequenceClassification.from_pretrained("bert-base-uncased"

for param in model1.bert.parameters():
    param.requires_grad = False
```

```
model.safetensors: 100%                                    440M/440M [00:06<00:00, 66.6MB/s]

Loading weights: 100%  199/199 [00:00<00:00, 630.88it/s, Materializing param=bert.pooler.dense.we

BertForSequenceClassification LOAD REPORT from: bert-base-uncased
Key                                          | Status     |
---------------------------------------------+------------+-
cls.predictions.bias                         | UNEXPECTED |
cls.predictions.transform.dense.bias         | UNEXPECTED |
cls.predictions.transform.LayerNorm.weight   | UNEXPECTED |
cls.predictions.transform.LayerNorm.bias     | UNEXPECTED |
cls.seq_relationship.weight                  | UNEXPECTED |
cls.predictions.transform.dense.weight       | UNEXPECTED |
cls.seq_relationship.bias                    | UNEXPECTED |
classifier.weight                            | MISSING    |
classifier.bias                              | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; r
- MISSING   :those params were newly initialized because missing from the checkp
```

```
training_args1 = TrainingArguments(
    output_dir="./results1",
    eval_strategy="epoch",
    learning_rate=2e-5,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    num_train_epochs=2,
    report_to="none"
)

trainer1 = Trainer(
    model=model1,
```

```
    args=training_args1,
    train_dataset=train_dataset,
    eval_dataset=val_dataset,
    data_collator=DataCollatorWithPadding(tokenizer),
    compute_metrics=compute_metrics
)

trainer1.train()
res1 = trainer1.evaluate(test_dataset)
print(res1)
```

[876/876 01:08, Epoch 2/2]

| Epoch | Training Loss | Validation Loss | Accuracy | Precision | Recall | F1 |
|---|---|---|---|---|---|---|
| 1 | No log | 0.693484 | 0.504000 | 0.503541 | 0.504000 | 0.501611 |
| 2 | 0.700102 | 0.691587 | 0.520667 | 0.521323 | 0.520667 | 0.510026 |

Writing model shards: 100% 1/1 [00:06<00:00, 6.90s/it]

Writing model shards: 100% 1/1 [00:01<00:00, 1.50s/it]

[94/94 00:05]

```
{'eval_loss': 0.6926450729370117, 'eval_accuracy': 0.5033333333333333, 'eval_pre
```

## Experiment 2: Fine-Tune Last Two Layers

In this experiment, the last two layers of BERT are fine-tuned along with the classifier. This allows the model to adapt better to the dataset and improves performance.

```
model2 = AutoModelForSequenceClassification.from_pretrained(
    "bert-base-uncased",
    num_labels=2
)

for param in model2.bert.parameters():
    param.requires_grad = False

for param in model2.bert.encoder.layer[-2:].parameters():
    param.requires_grad = True

for param in model2.classifier.parameters():
    param.requires_grad = True
```

```
Loading weights: 100%  199/199 [00:00<00:00, 630.84it/s, Materializing param=bert.pooler.dense.we

BertForSequenceClassification LOAD REPORT from: bert-base-uncased
Key                                          | Status     |
---------------------------------------------+------------+-
cls.predictions.bias                         | UNEXPECTED |
cls.predictions.transform.dense.bias         | UNEXPECTED |
cls.predictions.transform.LayerNorm.weight   | UNEXPECTED |
cls.predictions.transform.LayerNorm.bias     | UNEXPECTED |
cls.seq_relationship.weight                  | UNEXPECTED |
cls.predictions.transform.dense.weight       | UNEXPECTED |
cls.seq_relationship.bias                    | UNEXPECTED |
classifier.weight                            | MISSING    |
classifier.bias                              | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; r
- MISSING   :those params were newly initialized because missing from the checkp
```

```
training_args2 = TrainingArguments(
    output_dir="./results2",
    eval_strategy="epoch",
    learning_rate=2e-5,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    num_train_epochs=2,
    report_to="none"
)

trainer2 = Trainer(
    model=model2,
    args=training_args2,
    train_dataset=train_dataset,
    eval_dataset=val_dataset,
    data_collator=DataCollatorWithPadding(tokenizer),
    compute_metrics=compute_metrics
)

trainer2.train()
res2 = trainer2.evaluate(test_dataset)
print(res2)
```

[876/876 01:35, Epoch 2/2]

| Epoch | Training Loss | Validation Loss | Accuracy | Precision | Recall | F1 |
|---|---|---|---|---|---|---|
| 1 | No log | 0.432532 | 0.799333 | 0.799985 | 0.799333 | 0.799171 |
| 2 | 0.514900 | 0.422785 | 0.807333 | 0.807932 | 0.807333 | 0.807192 |

Writing model shards: 100% 1/1 [00:01<00:00, 1.44s/it]

Writing model shards: 100% 1/1 [00:08<00:00, 8.60s/it]

[94/94 00:05]

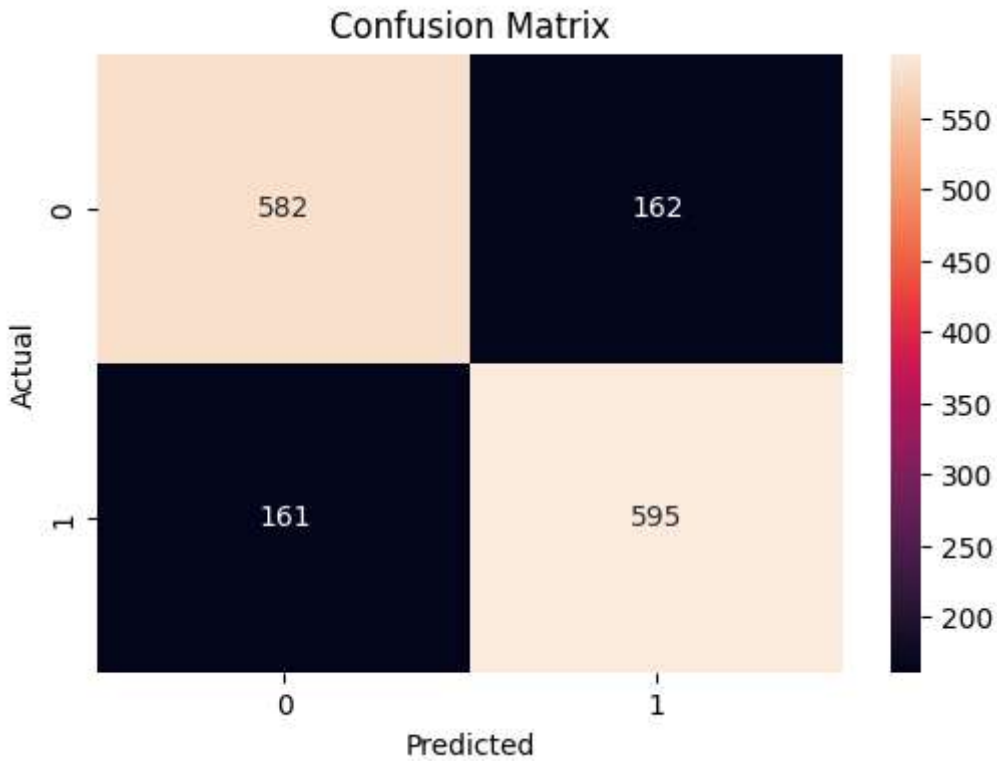
```
{'eval_loss': 0.4572555422782898, 'eval_accuracy': 0.7846666666666666, 'eval_pre
```

# CONFUSION MATRIX

```
preds = trainer2.predict(test_dataset)
y_pred = np.argmax(preds.predictions, axis=1)

cm = confusion_matrix(test_labels, y_pred)

plt.figure(figsize=(6,4))
sns.heatmap(cm, annot=True, fmt='d')
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```




## CLASSIFICATION REPORT

```
print(classification_report(test_labels, y_pred))
```

```
              precision    recall  f1-score   support

           0       0.78      0.78      0.78       744
           1       0.79      0.79      0.79       756

    accuracy                           0.78      1500
   macro avg       0.78      0.78      0.78      1500
weighted avg       0.78      0.78      0.78      1500
```

## COMPARISON TABLE

```
comparison = pd.DataFrame([
    ["Freeze BERT Layers", res1["eval_accuracy"], res1["eval_precision"], res1[
    ["Fine-Tune Last 2 Layers", res2["eval_accuracy"], res2["eval_precision"],
], columns=["Experiment", "Accuracy", "Precision", "Recall", "F1 Score"])

comparison
```

| | Experiment | Accuracy | Precision | Recall | F1 Score |
|---|---|---|---|---|---|
| 0 | Freeze BERT Layers | 0.503333 | 0.502378 | 0.503333 | 0.492102 |